

Replication Plan and Proposed Extension for *Does the Election of a Female Leader Clear the Way for More Women in Politics?*

David Graz, Saowalak Giada De Rossi, Thomas Jurkowitz & Yangchen Norbu Dru

March 28, 2026

Project logic. The study proceeds in two stages. First, it replicates the original regression-discontinuity (RD) design in close mixed-gender mayoral races, focusing on women candidates' *normalized rank improvement*. Second, it extends the design by asking whether any spillover from electing a female mayor reaches women from *other parties*, whether the cross-party effect differs from the copartisan effect, and whether treatment effects are heterogeneous across candidate and municipal contexts.

1. Replication plan

1. **Reconstruct the missing file for Table A18.** Rebuild *neighbor_females.dta* using geographic data to derive municipal contiguity and compute neighboring female mayor shares.
2. **Replicate tables and figures using the provided dataset.** Use the variables already available in the main dataset and follow the same sample restrictions and empirical setup as in the original paper. Reproduce Tables 1–9, Appendix Tables A1–A19, and all figures using identical RD specifications, bandwidth choices, and polynomial orders.
3. **Validate the results.** Compare replicated estimates with the original outputs and verify that coefficients, standard errors, and overall conclusions are consistent.

2. Proposed extension

The extension develops the following research question: *Do female mayors generate cross-party spillovers in the electoral performance of female candidates, and are these spillovers heterogeneous across candidates, parties, municipalities, and election timing?* This question is operationalized through 4 nested claims:

- **H3a - Contemporary cross-party spillover:** test whether a narrowly elected female mayor improves the electoral rank performance of women from other parties in the same election.
- **H3b - Cross-party versus copartisan contrast:** estimate an interaction model on the all-women sample to test whether any spillover extends beyond the mayor's own party.
- **H3c - Dynamic validation:** use lagged and next-election outcomes as validation exercises, but interpret them as broader evidence on women candidates rather than as strict cross-party tests unless a dynamic cross-party sample is explicitly reconstructed from the underlying candidate and mayor files.
- **H3d - Heterogeneous effects:** study heterogeneity inside the RD window with a forest-based module estimated on the all-women sample, while keeping RD benchmarks as the identification anchor.

Expected contribution. The project does not merely reproduce the original result; it turns the paper into a sharper test of whether female leadership generates spillovers beyond party boundaries. The most credible endpoint is a **RD-anchored heterogeneity design**: average cross-party effects are tested with transparent benchmarks, while any richer pattern is examined through disciplined heterogeneity analysis rather than through over-extended causal claims.

3. Data availability and suitability

Data asset	Use and suitability
Official replication package (114710-V1)	Available and appropriate for benchmark replication; contains the main Stata data files, the README, and enough structure to recover the paper's core variables.
Main candidate-level file + party merge file	Well suited for the contemporary extension because they contain mayor gender, RD margin terms, female candidate identifiers, rank-improvement outcomes, and party information needed to build all-women, cross-party, and copartisan samples.
Lagged and next-election files	Useful for placebo and persistence checks, but only partially suitable for a strict cross-party design because the <code>joint_party</code> information needed for dynamic cross-party classification is not cleanly preserved.

Assessment. The data are **sufficient for a credible replication and for the main contemporary extension**. The main empirical limitations are not data absence but **scope and measurement constraints**: (i) dynamic files support broader women-candidate validation more clearly than strict cross-party dynamics, and (ii) richer control sets can generate non-trivial sample loss. For that reason, the preferred interpretation should rely on the minimal and quadratic RD anchor, with enriched specifications and forest estimates used as robustness and heterogeneity layers.

Key project references: Baskaran and Hessami (2018), official replication package 114710-V1, *Readme-Original.pdf*, the project prompt used to frame the replication and extension, and *ChatGPT(5.4)*, *Claude*, and *Gemini*, which were used as support tools.